

STAC67: Regression Analysis

Lecture 8

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University of Toronto Scarborough

Review of Lecture 7

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Today: what each assumption buys us, how to check them, and what to do when one fails.

Diagnostics and remedial measures

What each result needs

Result	Assumptions used
$\hat{\beta}_0, \hat{\beta}_1$ unbiased	L
$Var(\hat{\beta}_1) = \sigma^2 / S_{xx}$, Gauss-Markov (BLUE)	L, I, E
t and F tests, confidence and prediction intervals	L, I, E, N

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- Even there, it matters less than it seems for $\hat{\beta}_1$: $\hat{\beta}_1 = \beta_1 + \sum_i k_i \epsilon_i$ is a weighted **sum of n errors**, so by the central limit theorem it is close to normal for moderate n , whatever the distribution of the ϵ_i . [see next slide]

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We cannot observe the errors ϵ_i to perform statistical inference on them, so we check the assumptions with their approximations instead, the residuals $e_i = Y_i - \hat{Y}_i$, mostly by plotting them.

Why the slope is nearly normal anyway (optional)

From Lecture 3, $\sum_i k_i = 0$ and $\sum_i k_i x_i = 1$, so

$$\hat{\beta}_1 = \sum_{i=1}^n k_i Y_i = \sum_{i=1}^n k_i (\beta_0 + \beta_1 x_i + \epsilon_i) = \beta_1 + \sum_{i=1}^n k_i \epsilon_i.$$

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$$\frac{\sum_i k_i \epsilon_i}{\sigma \sqrt{\sum_i k_i^2}} = \frac{\hat{\beta}_1 - \beta_1}{\sigma / \sqrt{S_{xx}}} \longrightarrow N(0, 1).$$

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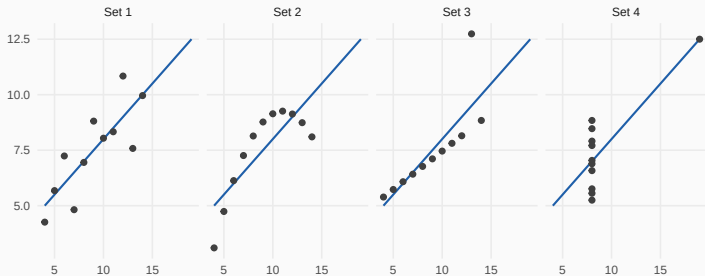
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The same holds for $\hat{\beta}_0$ and $\hat{Y}_{n+1}$.

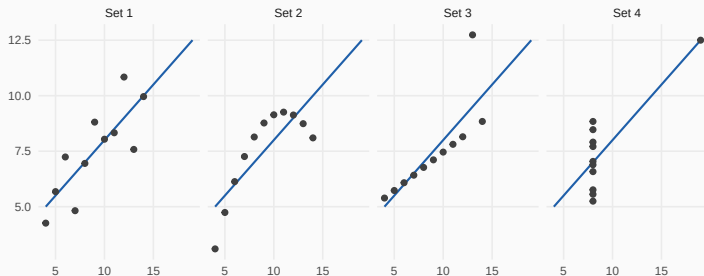
Checking assumptions

Reminder: in Lecture 2, we saw that the same r can correspond to different shapes



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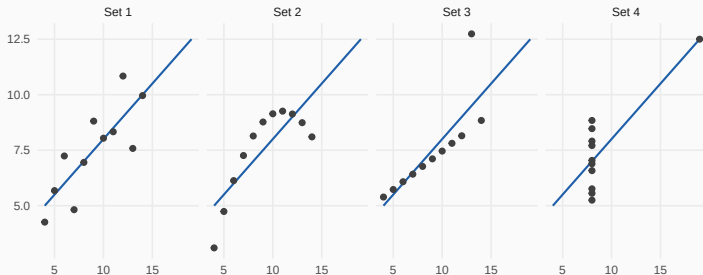
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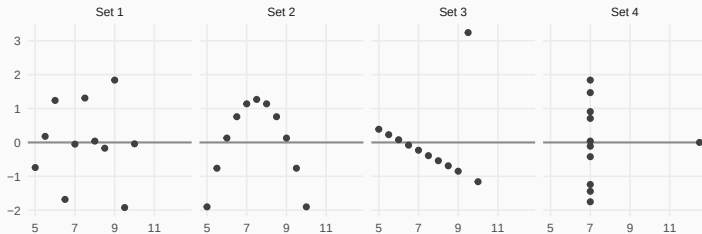


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The regression output cannot tell these four apart, but we can use a plot.

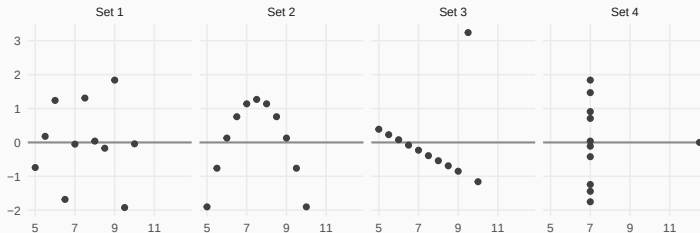
Residual plots

Residuals e_i plotted against the fitted values $\hat{Y}_i$:



Residual plots

Residuals e_i plotted against the fitted values $\hat{Y}_i$:



- Set 1: no pattern. This is what a residual plot should look like.
 - Set 2: a clear curve. **Linearity** fails.
 - Set 3: one outlier, while the other ten points lie exactly on a line.
 - Set 4: one point alone determines the slope, yet its residual is zero.
- ⚠️ A residual plot can miss a point with this much influence.

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- (E) The spread grows with the number of tests. A single σ for every day is too large for days with fewer tests, which is why the prediction intervals ran below zero.
- (I) Consecutive days resemble each other, so the errors are unlikely to be independent.

Semistudentized residuals: the residuals on the scale of σ ,

$$e_i^* = \frac{e_i}{\hat{\sigma}}, \quad \text{where } \hat{\sigma} = \sqrt{MSE}.$$

If the model fits the data well, we expect:

- the plot of these residuals against $\hat{Y}_i$ to look like a random scatter plot
- a histogram of the residuals to look normal
- a normal Q-Q plot of the residuals to lie close to a straight line
- no outliers: semistudentized residuals beyond about ± 3 deserve a second look

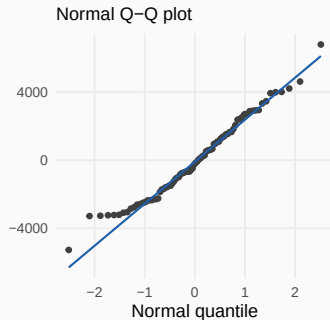
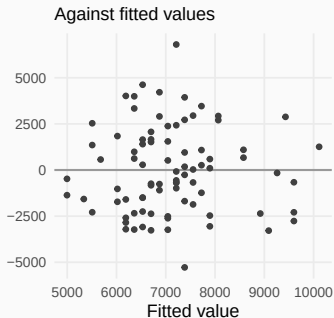
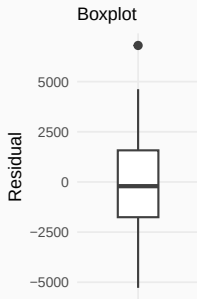
Exercise: residual analysis for the crime rate data

Using the crime rate data from Lectures 4 to 6:

1. Obtain the residuals e_i and draw a boxplot of them. Describe their distribution.
2. Plot the residuals e_i against the fitted values $\hat{Y}_i$. What does the plot show?
3. Draw a normal Q-Q plot of the residuals. Does normality look plausible?

In R: `resid(fit)`, `fitted(fit)`, and `stat_qq()` with `stat_qq_line()` in `ggplot2`.

The crime rate data: residual plots



A test for normality

The **Shapiro-Wilk test**, available in most statistical software:

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shapiro.test(resid(fit))
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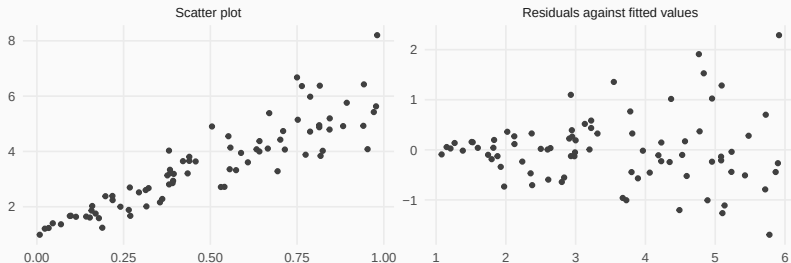
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 With a large n , the test rejects for departures from Gaussianity that are too small to matter (CLT). With a small n , it can miss real ones. The **Q-Q plot** remains the main tool.

Unequal variance: the trumpet shape



The spread of the residuals grows with the fitted value. A formal test:

H_0 : equal variance, $Var(\epsilon_i) = \sigma^2$ for all i

H_a : unequal variance, increasing or decreasing with X

The Brown-Forsythe test

1. Split the data into **two groups** by the level of X (or of $\hat{Y}$), of sizes n_1 and n_2 .
2. In each group, compute the median residual, $\tilde{e}_1$ and $\tilde{e}_2$.
3. Compute the absolute deviations from the group median: $d_{ij} = |e_{ij} - \tilde{e}_j|$.
4. Compute the mean and variance of the d_{ij} in each group: $\bar{d}_1, s_1^2$ and $\bar{d}_2, s_2^2$.
5. Pool the variances: $s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$.

The test statistic is

$$t_{BF} = \frac{\bar{d}_1 - \bar{d}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \stackrel{H_0}{\sim} t_{n-2}, \quad \text{reject } H_0 \text{ if } |t_{BF}| > t(1 - \alpha/2; n - 2).$$

Try it on the crime rate data, splitting at the median of X .

Remedial measures

- **Nonlinear relation**: add polynomial terms, fit an exponential regression function, or transform Y and/or X .
- **Non-constant variance**, $Var(\epsilon_i) = \sigma_i^2$: weighted least squares, or transform Y and/or X .
- **Dependent errors**: model the dependence, for instance with generalized least squares.
- **Non-normal errors**: transform Y , for instance with a Box-Cox transformation.
- **Outliers**: check the data first, then consider robust regression.

Several of these come back later in the course: polynomial regression, weighted least squares.

The Box-Cox transformation

Transform the response by a power, and choose the power from the data:

$$Y^* = \begin{cases} Y^\lambda, & \lambda \neq 0 \\ \log Y, & \lambda = 0 \end{cases} \quad (Y > 0)$$

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To compare values of λ on the same scale, use the standardized form

$$W_i = \begin{cases} \frac{Y_i^\lambda - 1}{\lambda K^{\lambda-1}}, & \lambda \neq 0 \\ K \log Y_i, & \lambda = 0 \end{cases} \quad K = \left(\prod_{i=1}^n Y_i \right)^{1/n},$$

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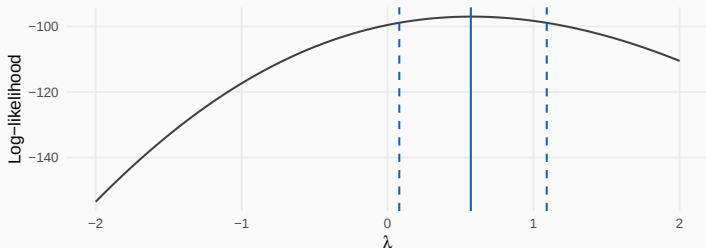
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The aim is normality, linearity and constant variance at once. It does not always succeed.

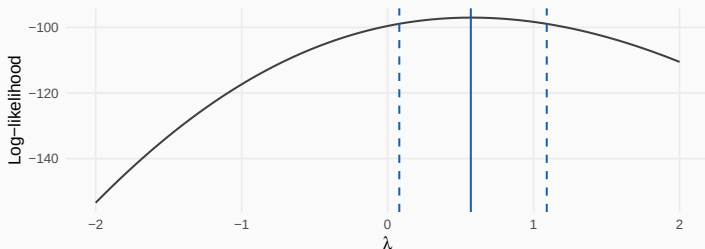
Box-Cox for the crime rate data

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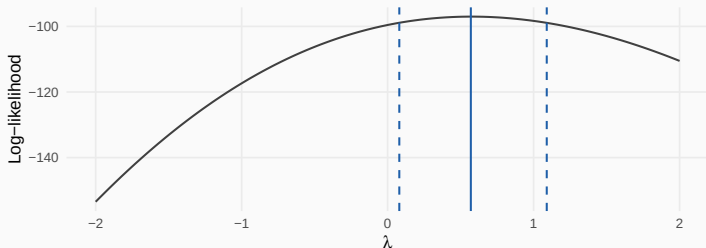
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The interval contains $\lambda = 1$: **the data do not require a transformation**. A square root, $\lambda = 1/2$, is equally plausible.

Lowess (smoothed) plots

A **lowess** curve gives a smooth summary of how Y varies with X , with no assumption on its shape:

- it fits a weighted regression in a small neighbourhood of each point, giving more weight to the closest observations;
- it refits, giving less weight to points with large residuals;
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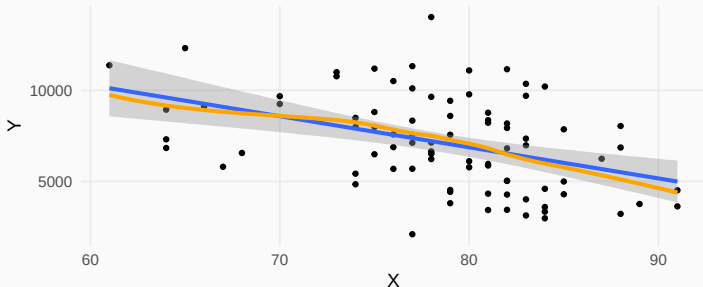
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Plotted with the linear fit and its confidence band: **if the lowess curve stays inside the band, the straight line is adequate.**

Lowess for the crime rate data

```
ggplot(Crime, aes(X, Y)) +  
  geom_point(size = 0.9) +  
  geom_smooth(method = "lm") +  
  geom_smooth(method = "loess", se = FALSE,  
             colour = "orange")
```



Blue: least squares line with its 95% confidence band. Orange: lowess.

Matrix approach to simple linear regression

Matrices

A **matrix** is a rectangular array of numbers or symbols. Its **dimension** $r \times c$ gives its number of rows and columns:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1c} \\ a_{21} & a_{22} & \dots & a_{2c} \\ \vdots & \vdots & & \vdots \\ a_{r1} & a_{r2} & \dots & a_{rc} \end{bmatrix} = [a_{ij}], \quad i = 1, \dots, r; \quad j = 1, \dots, c.$$

- **Square matrix**: as many rows as columns, $r = c$.
- **Vector**: a matrix with one column (column vector) or one row (row vector):

$$\mathbf{D} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}, \quad \mathbf{E}' = [17 \ 31], \quad \mathbf{F}' = [f_1 \ f_2 \ f_3].$$

Transpose

The **transpose** A' of a matrix swaps its rows and columns:

$$\mathbf{G}_{2 \times 3} = \begin{bmatrix} 6 & 15 & 22 \\ 8 & 13 & 25 \end{bmatrix}, \quad \mathbf{G}'_{3 \times 2} = \begin{bmatrix} 6 & 8 \\ 15 & 13 \\ 22 & 25 \end{bmatrix}.$$

Regression example. The **response vector** and the **design matrix**:

$$\mathbf{Y}_{n \times 1} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{X}_{n \times 2} = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}, \quad \mathbf{X}'_{2 \times n} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ X_1 & X_2 & \dots & X_n \end{bmatrix}.$$

Addition and subtraction

Add or subtract the corresponding elements. This is only defined for two matrices of the **same dimension**, and it is commutative: $A + B = B + A$.

Regression example. With the mean vector and the error vector

$$E(Y) = \begin{bmatrix} E(Y_1) \\ \vdots \\ E(Y_n) \end{bmatrix}, \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix},$$

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the model reads, one line for each observation:

$$Y = E(Y) + \epsilon \quad \text{since} \quad \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} E(Y_1) + \epsilon_1 \\ \vdots \\ E(Y_n) + \epsilon_n \end{bmatrix}.$$

Multiplication

By a scalar: multiply every element.

$$k = 3, \quad \mathbf{A} = \begin{bmatrix} 2 & 1 \\ -2 & 7 \end{bmatrix} \quad \Rightarrow \quad k\mathbf{A} = \begin{bmatrix} 6 & 3 \\ -6 & 21 \end{bmatrix}.$$

By a matrix: $\mathbf{AB}$ is defined when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$. Its (i, j) element combines row i of $\mathbf{A}$ with column j of $\mathbf{B}$:

$$\mathbf{A}_{r_A \times c} \mathbf{B}_{c \times c_B} = \left[\sum_{k=1}^c a_{ik} b_{kj} \right], \quad \begin{bmatrix} 2 & 5 \\ 3 & -1 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 16 & 18 \\ 7 & -7 \\ 14 & 28 \end{bmatrix}.$$

In general $\mathbf{AB} \neq \mathbf{BA}$, even when both products exist.

Multiplication in practice

Simultaneous equations. $a_{11}x_1 + a_{12}x_2 = y_1$ and $a_{21}x_1 + a_{22}x_2 = y_2$:

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$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad \text{that is, } \mathbf{Ax} = \mathbf{y}.$$

Multiplication in practice

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Regression function.

$$\mathbf{X}\boldsymbol{\beta} = \begin{bmatrix} 1 & X_1 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 X_1 \\ \vdots \\ \beta_0 + \beta_1 X_n \end{bmatrix} = \mathbf{E}(\mathbf{Y}).$$

Three products that come back all the time

$$\mathbf{Y}'\mathbf{Y} = \begin{bmatrix} Y_1 & \dots & Y_n \end{bmatrix} \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}$$

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Every sum in simple linear regression is hidden in these three products.

Special matrices

- **Symmetric**: a square matrix equal to its transpose, $A = A'$. For instance $X'X$.
- **Diagonal**: a square matrix whose off-diagonal elements are all 0.
- **Identity I** : diagonal, with 1 on the diagonal. $IA = AI = A$.

$$\mathbf{I}_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{1}_{r \times 1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \quad \mathbf{J}_{r \times r} = \begin{bmatrix} 1 & \dots & 1 \\ \vdots & & \vdots \\ 1 & \dots & 1 \end{bmatrix}, \quad \mathbf{0}_{r \times 1} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

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Note that $\mathbf{1}'\mathbf{1} = r$ and $\mathbf{1}\mathbf{1}' = \mathbf{J}$.

Linear dependence, rank and inverse

- The columns of a matrix are **linearly dependent** if a linear combination of them, with coefficients not all zero, gives the zero vector.
- The **rank** is the number of linearly independent columns (equivalently, rows): $\text{rank}(\mathbf{A}) \leq \min(r_A, c_A)$. The matrix has **full rank** if equality holds.

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For instance, in $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ the second column is twice the first: the rank is 1.

A **square, full rank** matrix $\mathbf{A}$ has an **inverse** $\mathbf{A}^{-1}$, such that

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}.$$

The inverse of a 2×2 matrix

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad |\mathbf{A}| = ad - bc, \quad \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If $\mathbf{A}$ is not of full rank, $|\mathbf{A}| = 0$ and there is no inverse.

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Regression example. $|\mathbf{X}'\mathbf{X}| = n \sum X_i^2 - (\sum X_i)^2 = nS_{xx}$, so

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{bmatrix} \sum X_i^2 & -\sum X_i \\ -\sum X_i & n \end{bmatrix}$$

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Multiply by σ^2 : the diagonal gives $Var(\hat{\beta}_0)$ and $Var(\hat{\beta}_1)$, and the off-diagonal $Cov(\hat{\beta}_0, \hat{\beta}_1)$, all from Lecture 4.

Useful matrix results

Addition

$$A + B = B + A, \quad (A + B) + C = A + (B + C)$$

Multiplication

$$(AB)C = A(BC), \quad C(A + B) = CA + CB, \quad k(A + B) = kA + kB$$

Transpose

$$(A')' = A, \quad (A + B)' = A' + B', \quad (AB)' = B'A'$$

Inverse (square, full rank matrices)

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (A^{-1})^{-1} = A, \quad (A')^{-1} = (A^{-1})'$$

The order reverses for transposes and inverses of products: $(ABC)' = C'B'A'$.

Determinant and trace

For square matrices of dimension $n \times n$:

Determinant:

$$|\mathbf{A}\mathbf{B}| = |\mathbf{A}| |\mathbf{B}|, \quad |\mathbf{A}'| = |\mathbf{A}|, \quad |\mathbf{A}^{-1}| = \frac{1}{|\mathbf{A}|}, \quad |k\mathbf{A}| = k^n |\mathbf{A}|$$

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The **trace** is the sum of the diagonal elements, $tr(\mathbf{A}) = \sum_{i=1}^n a_{ii}$.

Trace:

$$tr(\mathbf{A} + \mathbf{B}) = tr(\mathbf{A}) + tr(\mathbf{B}), \quad tr(k\mathbf{A}) = k tr(\mathbf{A}), \quad tr(\mathbf{A}') = tr(\mathbf{A})$$

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$$|AB| = |A| |B|, \quad |A'| = |A|, \quad |A^{-1}| = \frac{1}{|A|}, \quad |kA| = k^n |A|$$

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Trace:

$$tr(A + B) = tr(A) + tr(B), \quad tr(kA) = k tr(A), \quad tr(A') = tr(A)$$

and, above all, the trace is unchanged by moving a matrix from the front to the back:

$$tr(AB) = tr(BA), \quad tr(ABC) = tr(BCA) = tr(CAB).$$

Here AB and BA need only be square, not A and B themselves.

Random vectors

A vector of random variables, shown for $n = 3$:

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}, \quad \text{with mean} \quad \mathbf{E}(\mathbf{Y}) = \begin{bmatrix} E(Y_1) \\ E(Y_2) \\ E(Y_3) \end{bmatrix}.$$

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$$\sigma^2(\mathbf{Y}) = E [(\mathbf{Y} - \mathbf{E}(\mathbf{Y}))(\mathbf{Y} - \mathbf{E}(\mathbf{Y}))']$$

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where $\sigma_i^2 = \text{Var}(Y_i)$ and $\sigma_{ij} = \text{Cov}(Y_i, Y_j)$.

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Entry (i, j) is $E[(Y_i - EY_i)(Y_j - EY_j)] = \text{Cov}(Y_i, Y_j)$: the matrix is symmetric, with the variances on the diagonal.

Simple linear regression in matrix form

The model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$, for $i = 1, \dots, n$, becomes

$$\underset{n \times 1}{\mathbf{Y}} = \underset{n \times 2}{\mathbf{X}} \underset{2 \times 1}{\boldsymbol{\beta}} + \underset{n \times 1}{\boldsymbol{\epsilon}}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}.$$

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With independent errors of constant variance,

$$\sigma^2(\boldsymbol{\epsilon}) = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix} = \sigma^2 \mathbf{I}, \quad \text{and} \quad \sigma^2(\mathbf{Y}) = \sigma^2 \mathbf{I}.$$

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If the errors are also normal: $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$, and so $\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$.

Least squares in matrix form

Minimize

$$Q(\beta) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 = (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta).$$

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The normal equations of Lecture 3 become

$$\begin{bmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} \sum Y_i \\ \sum X_i Y_i \end{bmatrix}, \quad \text{that is,} \quad \mathbf{X}'\mathbf{X} \mathbf{b} = \mathbf{X}'\mathbf{Y}.$$

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If $\mathbf{X}'\mathbf{X}$ is invertible (full rank), the solution is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}.$$

Checking against Lecture 3

With $(\mathbf{X}'\mathbf{X})^{-1}$ from the 2×2 inverse,

$$\hat{\beta} = \frac{1}{nS_{xx}} \begin{bmatrix} \sum X_i^2 & -\sum X_i \\ -\sum X_i & n \end{bmatrix} \begin{bmatrix} \sum Y_i \\ \sum X_i Y_i \end{bmatrix}.$$

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Second row:

$$\hat{\beta}_1 = \frac{n \sum X_i Y_i - \sum X_i \sum Y_i}{nS_{xx}} = \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{S_{xx}} = \frac{S_{xY}}{S_{xx}}.$$

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First row:

$$\hat{\beta}_0 = \frac{\sum X_i^2 \sum Y_i - \sum X_i \sum X_i Y_i}{nS_{xx}} = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

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The same estimates, in a form that extends unchanged to several predictors.